Lab 4: VibeCoding — Minesweeper

Focused conversation extract and testing record

Scope note: This is a curated, focused extract of the Minesweeper-related work from the ChatGPT conversation. It is not a verbatim export of every message. The original conversation link is retained separately for access to the full history.

1. Repository and task

Assigned starter repository: SETAPESU26/13_minesweeper
Personal fork: cswizard7/13_minesweeper
Local working folder: `C:\Users\shree\Shreesh\PES\sem 5\SE\13_minesweeper`

The work focused on the README tasks: boundary-safe neighbor traversal; win/flag behaviour; difficulty modes; and one action-level feedback message per reveal. Changes were made in the existing modular code rather than replacing the project.

2. Bug identified and fix

In `board.py`, the neighbor boundary condition allowed row/column indices equal to the board dimensions, producing out-of-bounds neighbors. The condition was changed from inclusive upper bounds to exclusive upper bounds.

```
Original:
if 0 <= nr <= self.rows and 0 <= nc <= self.cols:

Updated:
if 0 <= nr < self.rows and 0 <= nc < self.cols:
```

The diagnostic test after the fix reported zero invalid neighbors. Further checks reported: corner has 3 neighbors, edge has 5 neighbors, center has 8 neighbors, zero-adjacent flood-fill reveals the board, and the win condition is detected after all safe cells are revealed.

3. Feature changes

Difficulty selection: Added Easy, Medium, and Hard options in `game.py`.

Mode	Dimensions	Mine count
Easy	6 × 6	6
Medium	8 × 8	12
Hard	10 × 10	20

Reveal feedback: A safe reveal prints `Cell revealed.` once per player reveal action, outside the flood-fill loop. A repeated reveal prints `Cell already revealed.`. Mine hits show `BOOM! You hit a mine.` and end the game. Existing flag toggling was tested, including the rule that a revealed cell cannot be flagged.

4. Prompts and problem-solving workflow

The interaction followed an incremental workflow: inspect the existing files and README; identify the boundary defect; make a minimal fix; add requested features one at a time; run focused tests; and verify syntax and Git changes before committing. The assistant provided step-by-step CMD instructions, and the user ran commands and reported the actual outputs.

Representative task prompts (summarized, not quoted verbatim): fix neighbor traversal so only valid coordinates are returned; check win detection and flag behaviour; add three difficulty configurations without dependencies; add one reveal-feedback message and repeated-reveal handling; test incrementally and review the final diff before committing.

5. Testing evidence

The following commands were run by the user in Windows CMD. No error output was reported for either final check.

```
python -m py_compile board.py game.py main.py
git diff --check
```

Mine-hit test command and observed output:

```
python -c "from board import Board; b=Board(rows=3,cols=3,mines=1); b.mines={(1,1)}; b.mine_total=1; hit=b.reveal((1,1)); asser

PASS: mine hit detected
PASS: mine cell revealed
```

Win-condition test output:

```
Before revealing safe cells: PASS
After revealing all safe cells: PASS
```

Other checks performed: invalid-neighbor count was 0; corner/edge/center neighbor counts were correct; zero-region flood-fill and win detection were tested; Easy/Medium/Hard and invalid difficulty input were tested; flag add/remove and flagged-cell reveal protection were tested; repeated reveals produced 'Cell already revealed.'; and gameplay showed 'Cell revealed.'

6. Evidence and version control

Before and after gameplay videos were saved as `before.mp4` and `after.mp4`. A screenshot of the original bug was saved as `before_bug_evidence.png`. These files were copied to the Lab-4 submission folder.

Only `board.py` and `game.py` were staged and committed. Commit: `31200f0`, message: “Fix Minesweeper bugs and add difficulty modes”. The push succeeded to the personal fork's `main` branch.

Repository: https://github.com/cswizard7/13_minesweeper

7. Submission files

• before.mp4 — original behaviour
• after.mp4 — updated behaviour
• before_bug_evidence.png — original bug evidence
• board.py, game.py, main.py — project source
• requirements.txt and README.md — setup and task description
• Lab4_VibeCoding_Chat_Extract.pdf — this focused record
• Original ChatGPT conversation link — full context (stored separately by the student)

Integrity note: This document summarizes the relevant interaction and includes selected actual command outputs. It is not a full transcript and should be submitted as a focused extract. Use the original conversation link for the complete history.